

$$T_{E_n} = U + U(2) + E_8(2) = (12, 10, 0)_2 \quad S_{E_n} = U(2) + E_8(2) = (10, 10, 0)_2$$

$$T_{C_0} = \langle 2 \rangle + U(2) + E_8(2) = (11, 11, 1)_2$$

$$= (\langle 1 \rangle + U + E_8)(2) = I_{2,9}(2)$$

$$\text{cusps of } I_{2,9}(2) \xleftrightarrow{1:1} \text{cusps of } I_{2,9}$$

$$i_1 I_{2,9}(2) \left\{ \begin{array}{l} e^2 = 0 \\ e_1 \cdot e_2 = 0 \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} e^2 = 0 \\ e_1 \cdot e_2 = 0 \end{array} \right\} i_1 I_{2,9}$$

Cusps of  $I_{2,9}$ :

$$\begin{array}{l} \text{0-cusps} \quad e \in I_{2,9} \quad e^2 = 0 \quad e^\perp/e \text{ unimod. sign}(1,8) \\ \exists! \text{ such } I_{1,8} \end{array}$$

$$\text{1-cusps of } I_{2,9} \xleftrightarrow{\quad} \text{0-cusps of } I_{1,8} \ni e' \quad e'^2 = 0$$

$$e'^\perp/e' \text{ unimod. sign}(0,7) \Rightarrow \text{odd unimod.} \Rightarrow I_{0,7} \leftarrow (-1)^{\oplus 7}$$

$$\begin{array}{l} \text{For } I_{2,9}(2): \quad \exists! \text{ 0-cusp } I_{1,8}(2) = (9, 9, 1)_1 = e^\perp/e \\ \exists! \text{ 1-cusp } I_{0,7}(2) = (7, 7, 1)_0 = J^\perp/J \end{array}$$

$$\begin{array}{ccccc} (7, 7, 1)_0 & \leftarrow & (9, 9, 1)_1 & \leftarrow & (11, 11, 1)_2 = T_{C_0} \\ J^\perp/J & & e^\perp/e & & \end{array}$$

$$\text{Maps } (\text{cusps of } T_{C_0}) \xrightarrow[\substack{e^2=0 \\ e}]{\substack{e^2=0 \\ e}} (\text{cusps of } T_{E_n})$$

0-cusps

$$\begin{array}{l} T_{C_0} \hookrightarrow T_{E_n} \text{ isometry: preserves bil. form} \\ \cup \\ J \\ \text{isomorphism: } \forall x, y \in J \quad x \cdot y = 0 \\ i(J) \text{ also an iso plane} \\ i(J)^{\text{set}} = \text{the corr. 1-cusp} \end{array}$$

$\alpha_1^* \dots \alpha_8^*$  weights

$e, f$   $e', f'$   $\alpha_1 \dots \alpha_8$  roots

$$T_{E_8} = U + U(2) + E_8(2) = (12, 10, 0)_2$$

$e+f$

$\uparrow$

$h$

$$T_{C_0} = \langle 2 \rangle + U(2) + E_8(2) = (11, 11, 1)_2$$

$$e' \in T_{C_0}, \quad e'^2 = 0 \quad e' \mapsto e'$$

$$\text{div}_{T_{E_8}} e' = 2$$

$$U(2): \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

$$(0, e', 0) \in U \oplus U(2) \oplus E_8(2) \Rightarrow e' \text{ is even}$$

$$(x, y, z)$$

$$\text{product} = 0 \cdot x + e' \cdot y + 0 \cdot z = e' \cdot y \in 2\mathbb{Z}$$

$$\text{In } T_{E_8}: \quad e'^{\perp}/e' \text{ has } (r, a, \delta) = (r-2, a-2, \delta')$$

$$e'^{\perp}/e' = (10, 8, 0) \text{ or } (10, 8, 1), \quad = (10, 8, \delta'),$$

$$U \oplus E_8(2) \quad U \oplus I_{0,8}(2) \quad L^* \ni x$$

$$\delta' = 0 \text{ or } 1$$

$$q(x) = x^2 \pmod{2\mathbb{Z}}$$

$$\forall \text{ 2-el. lattice } L = (r, a, \delta) \quad A_L = L^*/L \simeq \mathbb{Z}_2^a \ni \bar{e}$$

$$e, e^2 = 0 \quad \frac{e}{\text{div}(e)} = \bar{e} \quad \bar{e}^2 = 0 \text{ in } (A_L, q_L)$$

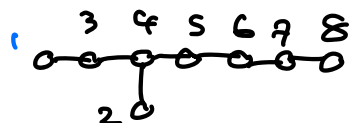
$$A_{e^{\perp}/e} = \bar{e}^{\perp}/\bar{e} \simeq \mathbb{Z}_2^{\dots}$$

Let's compute directly:  $U + U(2) + E_8(2)$

$$\begin{pmatrix} 0 & e' & 0 \\ x & y & z \end{pmatrix}$$

$$e'^{\perp} = U + \langle e' \rangle + E_8(2)$$

$$e'^\perp / e' = \underset{\substack{\uparrow \\ \delta=0}}{U} + \underset{\substack{\uparrow \\ \delta=0}}{E_8(2)} = (10, 8, 0)_1$$



1- cases:  $J \subset T_{C_0}$   $J = \langle e', 2h + \alpha_1 + \alpha_2 \rangle$   
 isotropic plane

$$(2h + \alpha_1 + \alpha_2)^2 = 8 - 4 - 4$$

$$\alpha_i^2 = -4 \quad \text{in } E_8(2)$$

$$T_{E_8} \supset i(J) = \langle e', 2e + 2f + \alpha_1 + \alpha_2 \rangle = i(J)^{\text{set}}$$

$$U + U(2) + E_8(2)$$

If  $a_1, a_2 \in \mathbb{Z}$   $(a_1, a_2) = 1$

$\Rightarrow a_1 e' + a_2 (2e + 2f + \alpha_1 + \alpha_2)$  is primitive

in  $U + U(2) + E_8(2)$

$$\frac{i(J)^\perp}{i(J)} = \frac{(2e + 2f + \alpha_1 + \alpha_2)^\perp}{\langle 2e + 2f + \alpha_1 + \alpha_2 \rangle} \quad \text{in}$$

$$e'^\perp / e' = \underset{\substack{\uparrow \\ \delta=0}}{U} + \underset{\substack{\uparrow \\ \delta=0}}{E_8(2)} = (10, 8, 0)_1$$

also  $(2e + 2f + \alpha_1 + \alpha_2) \leq 2$

$$(x, y, z) = \underbrace{2x + 2y}_{\text{even}} + \underbrace{(\alpha_1 + \alpha_2)z}_{\text{even}}$$

$$v^\perp / v = (8, 6, 0)_0$$

$\forall$  0-cusp we have  
 an  $IAS^2$ , Kulikov,  
 $KSDA$  stable models



(cusps of  $T_{E_n}$   
 mod  $\Gamma$ )



(cusps mod  
 $\Gamma'$ )

Need:

$T_{E_n} \hookrightarrow \Gamma$   
 $\cup$

$T_{C_0} \hookrightarrow \Gamma' = \text{Stab}_{T_{C_0}} \Gamma$